Does the daily video coach better when its prompt asks for a coach's moves?

One bullet added to the daily routine's prompt, replayed on 19 past days

Chess Coach research lab

Abstract

The app's daily routine writes a narrated video of the player's games every day. A titled coach's video review of a player near his level coaches with five moves: ask his move first, name the habit behind the mistake and the habit that replaces it, play the better move on to the opponent's best reply, say whether the miss is normal at his level, and praise a good move by what it saw. The routine's videos make three of these rarely or never. We added one bullet asking for the five moves to the routine's prompt and replayed the video step on 19 past days, once under each prompt, with the routine's own model and checker. Judging blind in both orders, a model preferred the candidate's coaching on 16 days and today's prompt's on none, on the train days and the held-out days alike, and preferred the candidate overall on 13 days against two. An audit against the evidence found the candidate asking his move first in 74% of its moment scenes (today's prompt: 8%), playing the better move on to their reply in 74% (21%) and placing the miss at his level in 83% of the positions it teaches (today's prompt: 0%), with no more false claims. The cost is length: its videos are 1.23 times as long heard once, and the judge found them harder to follow at speaking pace on eight days. Every condition of the rule fixed before the first run held, and the routine's prompt took the bullet.

1 The question

The app's daily routine has a model review the player's games of the day and write, besides the written review, a narrated video script: scenes with a board from one of his games, a chart or tiles, and beats with a spoken line, a caption, the moves played on the board and a pause where he is meant to think. The app renders it with boards, captions and a synthetic voice.

A titled player's video review of the rapid games of an adult rated about 1100, *Anyone Can Be 1200 In Chess. Watch This.* (`https://www.youtube.com/watch?v=PcGDehRSqTM`; the lab read a summary of its captions, not the video itself), coaches with five moves:

1. ask before telling: the student says what he would play before any hint;
2. name the habit behind the mistake, then the habit that replaces it;
3. "and then what?": follow the better move to the opponent's best reply;
4. pitch each miss at the player's level: a miss most players at his rating make is a pattern to learn, one they find is his to find;
5. praise a good move by what it saw.

The app's video grammar already asks for a pause to predict in every moment, findability or mirror framing, and a rule card at the end of each moment, but the routine's prompt says only how a video binds its boards and numbers. An audit of the 19 published scripts of the days below (Table 2)

found the second move in most moment scenes and the first, third and fourth rarely. The candidate adds one bullet of five sub-points to step 4 of the daily prompt, right after the bullet on how the video binds its boards:

> the video coaches the way a strong coach works with a player at his level, in every moment segment, from the evidence alone and inside the video grammar's chapters and rules:
>
> - ask before you tell: the segment's first pause asks what he would play there, open ("What would you play here? Pick two moves."), before any question that narrows it to a kind of move and before any arrow;
> - name the habit, then its replacement: say what kind of move he played and how (its scan step, the seconds it got, what it left behind, as the digest's `your move` and `it allowed` lines say), and show how the segment's rule card replaces that habit at the board;
> - and then what: after the better move, play or say their best reply from the digest's engine line and why the move still holds, so the line stops where its gain is clear;
> - pitch it at his level: say whether this is a miss most players at his rating make or one he could have found, from how often players at his band play the best move there (the `find` the Maia step wrote on that turning point in tools/coach-review/.cache/analysis.json, or `npm run -silent coach - engine find <fen> <uci> -rapid <his rating>`), in words and never as a percentage (under about 15%: a pattern to learn; half or more: his to find); on a hung piece, the mirror instead, as the video grammar says;
> - praise with a board: on a move he got right, name what it saw.

Nothing else in the prompt changes. The question is whether the video coaches better with it, without becoming less true or much longer.

2 The cases and the writer

A case is a past day whose review landed on the app's main branch with a video script: 23 days from September 25 to October 19. Each is a checkout of the app with the coach data as main held it just before that day's review landed and the day's archived engine analysis; the app's own `coach daily prepare` wrote the day's brief again, offline. The day's annotations, coach log, briefing and note replies are put back as they landed, so only the video script is missing, and a writer run writes the video for the review the day published. A case is usable only when the published video script passes today's checker with those files; September 26, October 8, October 10 and October 11 do not, which leaves 19 days with one to 14 games each. Most cases run the app at `3a377486e8` (13 days); the lab's pin moved while the cases were being built, so Oct 12, Oct 13, Oct 14, Oct 15, Oct 17 and Oct 19 run at `3ff95c4b72`, which adds the lesson lines' naming of pins (chess PR 136) and changes outside the routine. The prompt and the video checker are the same at both, and each day's two prompts ran on the same checkout. The split was fixed before any run, interleaved by date: train is Sep 25, Sep 27, Sep 29, Oct 1, Oct 4, Oct 6, Oct 12, Oct 14, Oct 17 and Oct 19; held out is Sep 28, Sep 30, Oct 2, Oct 5, Oct 7, Oct 9, Oct 13, Oct 15 and Oct 18.

The writer runs as the routine does: Opus 5.5, headless, with nothing inherited from the session that launched it, and a short preamble that names the day, says which steps ran and asks for the video script alone, done when the routine's check passes. Each prompt wrote every day once (44 runs in all), and today's prompt wrote Sep 25, Sep 27, Sep 28, Sep 29, Sep 30 and Oct 2 a second time, to measure the judge's noise floor.

3 The graders

A script with no model reads each run: whether the final check passes, the check rounds, files changed besides the video script, commands it should not have run, the script's shape, its length heard once (spoken words at 2.6 a second plus every pause), cost and minutes.

Sonnet 5.5 audits each script alone against the evidence it was written from: the brief without its bulk, the clock's turning points, a line per game, the engine digest of the day, how often players at his level play the engine's best move at each turning point, and for every board scene the position as the archive replays it, what was played there and next, the engine's best move and reply, how players at his band play there (Maia-2), the engine's plan lines, the threats, what each piece hits and guards, what each move the scene plays does, and after each move the game did not play, the engine's best reply. It lists the claims the evidence contradicts (*false*), does not carry (*unsupported*) or carries but slants (*misleading*), and for each board scene its kind (a moment, praise, a judgement call, a plan, other), its false claims about the board, and which of the five moves it makes.

Sonnet 5.5 also judges each pair blind, in both orders, on four criteria: truth, coaching, listening (would he follow it once, at speaking pace) and overall. A side wins a criterion only when both orders agree. Two controls check the judge on every day: today's script against itself, and against a decoy that adds as many spoken words as the candidate added that day (at least 40), in beats that restate their scene with no square, move, piece, number or question the scene does not already have.

The graders' prompts were settled on published scripts alone, before either prompt's runs were graded. The pilot (the first three train days) then showed two faults, and both were fixed before anything past the pilot was graded: the evidence did not carry what the writer's tools show (the Maia-2 reading, the engine's reply to a move the game did not play, the plan lines), so the audit graded claims the writer had read from them as unsupported; and one audit filed most of the lines it checked and found true as misleading, so the claim list gained a grade for a line that holds, which counts for nothing. Every script was graded again with the final graders. Graded with the first evidence, the judge preferred the candidate on coaching on September 25; the regrade's verdicts are the ones reported.

4 The rule

A day scores $+1$ when the candidate wins a criterion, -1 when it loses and 0 for a tie; intervals are 95% percentile bootstrap intervals over days. The candidate replaces the live prompt only when every condition of Table 1 holds, fixed before its first run.

Condition	Estimate (95% interval)	Holds
Coaching on the train days above 0	+0.80 (+0.50 to +1.00)	yes
Coaching on the held-out days above 0	+0.89 (+0.67 to +1.00)	yes
Overall not below 0	+0.58 (+0.26 to +0.90)	yes
False claims per script not up	+0.05 (−0.32 to +0.47)	yes
False board claims per script not up	+0.32 (−0.21 to +0.90)	yes
The check passes as often	19 of 19 (today's: 19)	yes
Controls clean (noise floor: coaching)	+0.17 (−0.33 to +0.67)	yes
Coaching moves made more often	+0.26 (+0.18 to +0.34)	yes
At most 1.3 times as long	1.23 (1.16 to 1.29)	yes

Table 1: The rule, fixed before the candidate's first run. Coaching and overall are the judge's scores per day; the claim rows and the coaching moves are the audit's numbers per script, candidate minus today's prompt; the length is the candidate's script heard once over today's on the same day.

5 What the bullet changed

In the moment scenes	Published	Today's prompt	Candidate
Asks what he would play before any hint	6% (65)	8% (77)	74% (82)
Names the habit behind the mistake	80% (74)	87% (77)	69% (120)
Ends on the habit that replaces it	97% (74)	92% (77)	65% (120)
Plays out their best reply to the better move	20% (74)	21% (72)	74% (77)
Says whether the miss is normal at his level	0% (74)	0% (77)	57% (120)
and that agrees with the evidence	–	–	91% (68)
Counted once per position taught			
Names the habit behind the mistake	80% (74)	87% (77)	89% (81)
Ends on the habit that replaces it	97% (74)	92% (77)	85% (81)
Says whether the miss is normal at his level	0% (74)	0% (77)	83% (81)
Praise names what the good move saw	69% (26)	81% (31)	86% (29)
Any board scene: an open question first	31% (127)	28% (185)	56% (185)

Table 2: The coaching moves the audit finds, pooled over the 19 days: the share of the scenes that owe a move and make it, with the number owing it in brackets. A moment scene owes the habit, its replacement and the level; it owes the open question only when it asks him something, and the reply only when it shows a better move. In the middle rows, consecutive board scenes on the same position count once, and a move counts as made when any of them makes it.

The published scripts and today's prompt name the habit behind a mistake and end on its replacement in most moment scenes, but they rarely ask his move before giving a hint (6% and 8%), rarely play the better move on to the opponent's best reply (20% and 21%), and never say whether a miss is normal at his level. The candidate does all three in most moment scenes (74%, 74% and 57%), and the level it states agrees with the findability evidence in 91% of the scenes that state one. Over the five moves, the share the audit finds in the moment scenes rose from 43% to 69% (+0.26 (+0.18 to +0.34) a script; rule 7).

Scene by scene, the candidate names the habit and its replacement less often (69% and 65%). That is how it asks first: it often shows a position in two scenes, one that asks and one that answers (2.2 board scenes a script repeat the position of the board scene before them, against 0.2 under today's

prompt), and the audit owes the habit, its replacement and the level on each moment scene, so the scene that asks counts as a miss. Counted once per position taught, the candidate names the habit as often as today's prompt (89% against 87%) and ends on its replacement a little less often (85% against 92%). This count was made after the result and decides nothing.

6 What the judge preferred

Criterion	Train (ten days)	Held out (nine days)	All
Truth	+0.10 (−0.20 to +0.40)	−0.11 (−0.44 to +0.22)	0.00 (−0.26 to +0.26)
Coaching	+0.80 (+0.50 to +1.00)	+0.89 (+0.67 to +1.00)	+0.84 (+0.68 to +1.00)
Listening	−0.30 (−0.60 to 0.00)	−0.56 (−0.89 to −0.22)	−0.42 (−0.63 to −0.21)
Overall	+0.40 (0.00 to +0.80)	+0.78 (+0.33 to +1.00)	+0.58 (+0.26 to +0.90)

Table 3: The judge's mean score per day from the candidate's side (+1 a win, −1 a loss, 0 a tie), with its 95% bootstrap interval over days.

The judge preferred the candidate's coaching on 16 of the 19 days, tied on three and never preferred today's prompt's, and it preferred the candidate overall on 13 days against two (Table 3). Truth was even: three wins, three losses and 13 ties. Listening went the other way: the candidate never won it and lost it on eight days. Its videos ask more and run longer, and the judge found them harder to follow once at speaking pace.

Two runs of today's prompt on the same day, judged against each other, split on coaching on three of six days, each way at least once, for a mean of +0.17 (−0.33 to +0.67). The judge does pick a side between two scripts of one prompt, but not one way on every day it decides, as it did for the candidate. Read in one order only, the judge favoured the side it read second in 60% of its 168 decided verdicts, so a verdict that only one order gives is a tie.

7 The controls

Today's script judged against itself tied on every criterion on all 19 days. The decoy, today's script with on average 265 spoken words added that only restate their scene, never won coaching and lost listening and overall on all 19 days. The judge does not reward words that say nothing new, so the candidate's wins come from what its extra words say.

8 Truth

The audit found about as many false claims in the candidate's scripts as in today's (0.53 and 0.47 a script; the difference is +0.05 (−0.32 to +0.47)) and a few more false claims about the board (1.11 against 0.79 a script; +0.32 (−0.21 to +0.90)), within the noise; the candidate shows more board scenes (14.7 against 12.5 a script). Unsupported and misleading lines were about as common under both prompts (2.0 and 1.8 unsupported, 2.6 and 2.7 misleading a script).

9 Length, cost and the check

Every run of both prompts passed the routine's check (19 and 19 of 19). Heard once, the candidate's videos last 10.9 minutes against 9.0, with 1448 spoken words against 1192 and 15.0 questions against 9.7; on the same day it is 1.23 (1.16 to 1.29) times as long (rule 8 allows 1.3). The writer took 3.1 check rounds against 2.5, 10 minutes against 8 and $3.17 against $2.61 a day at list price.

10 The graders measured twice

The audit is noisy itself. The six pilot scripts audited twice with the same evidence (the second time with a claim grade for a line that holds, which asks nothing new of the scene tallies) moved by 6 points of coaching share on average and by up to 20, so one day's share in Table 4 says little, while the pooled difference (+0.26 (+0.18 to +0.34)) is well beyond that noise.

			Judge, from the candidate's side				Length	Coaching share	
Day	Split	Games	Truth	Coaching	Listening	Overall	ratio	Today	Candidate
Sep 25	train	5	0	0	0	0	1.10	56	61
Sep 27	train	4	0	+	0	+	1.13	40	71
Sep 28	held out	3	0	+	0	+	1.59	27	69
Sep 29	train	7	0	+	−	+	1.27	45	54
Sep 30	held out	6	−	+	0	+	1.05	47	67
Oct 1	train	3	0	+	0	0	1.34	50	74
Oct 2	held out	4	0	+	−	+	1.31	53	68
Oct 4	train	4	+	+	−	+	1.44	44	66
Oct 5	held out	9	0	+	−	+	1.14	23	37
Oct 6	train	13	0	+	0	+	1.03	44	55
Oct 7	held out	1	0	+	−	+	1.19	60	56
Oct 9	held out	4	+	+	−	+	1.28	47	65
Oct 12	train	4	0	0	0	0	1.07	40	77
Oct 13	held out	14	0	+	−	+	1.49	36	66
Oct 14	train	6	0	+	0	0	1.18	33	67
Oct 15	held out	8	0	+	0	+	1.21	40	100
Oct 17	train	5	+	+	0	+	1.23	40	63
Oct 18	held out	1	−	0	0	−	0.99	30	100
Oct 19	train	2	−	+	−	−	1.23	53	88

Table 4: Each day: the judge's verdict from the candidate's side (+ a win, − a loss, 0 a tie), the candidate's length over today's heard once, and the share of the coaching moves the audit finds in the moment scenes.

11 What the app takes

The candidate passed every condition of the rule (Table 1), so the daily routine's prompt took the bullet as it was tested (chess PR 161). Listening is its cost: the videos are longer, ask more, and were harder to follow once on eight days. A next round could ask for the same moves in fewer words, for instance one scene per moment that asks, waits and answers.

12 Limits

- The five moves are one change: the result shows they help together, not which of them does.

- A replay writes the video alone, after the day's review is in place; the routine writes it at the end of a long day's work. The published videos are shorter (5.9 minutes heard once) than either prompt's replays, so the length and its listening cost may differ live. Both prompts faced the same setting on every day.
- The graders are models. The judge picks a side between two runs of one prompt on some days and favours the side it reads second; the audit's tallies move between two readings. The controls, the noise floor and both orders bound these, and the graders are a different model from the writer.
- The level a script states is checked against the same findability the writer reads (Maia-2 at his band), so the audit checks that the writer read it right, not that Maia-2 knows his level.
- The cases ran at two app commits; the prompt and the checker are the same at both, and each day's two prompts ran on the same checkout.
- The eval spent $205 at list price ($125 on the writer, $79 on the graders), on the owner's subscription, whose seven-day window it moved from 39% to 53%.